

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 11A.3

DETERMINING ORDERS OF REACTION

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A.3: Determining Orders of Reaction
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	7 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 11A.3

Q1. Q1. Explain how a concentration–time graph can be used to show that a reaction is first order.

[2 Marks]

WCH14/01/Jan22/Q3(a)

Q2. Q2. A first order reaction has a rate constant of $2.5 \times 10^{-3} \text{ s}^{-1}$. Calculate its half-life.

[2 Marks]

WCH14/01/Jan22/Q3(b)

Q3. Q3. Describe how the initial rate of reaction can be found from a concentration–time graph.

[2 Marks]

WCH14/01/Oct21/Q4(a)

Q4. Q4. In an iodine clock reaction, the time taken for a color change is measured. State the assumption made when calculating the initial rate as $1/t$.

[1 Marks]

WCH14/01/Oct21/Q4(b)

SECTION B: WORKED MARK SCHEME — 11A.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Q1. Explain how a concentration-time graph can be used to show that a reaction is first order. • Measure successive half-lives (1) • Half-life is constant / independent of concentration (1)	[2]
Q2	Q2: Q2. A first order reaction has a rate constant of $2.5 \times 10^{-3} \text{ s}^{-1}$. Calculate its half-life. • $t_{1/2} = \ln 2 / k$ (1) • $t_{1/2} = 0.693 / 2.5 \times 10^{-3} = 277 \text{ s}$ (1)	[2]
Q3	Q3: Q3. Describe how the initial rate of reaction can be found from a concentration-time graph. • Draw a tangent to the curve at time $t=0$ (1) • Calculate the gradient of the tangent (1)	[2]
Q4	Q4: Q4. In an iodine clock reaction, the time taken for a color change is measured. State the assumption made when calculating the initial rate as $1/t$. • The rate is constant over the time measured / concentration changes are small (1)	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 11A.3**FAQ #1: Why is drawing a tangent at $t=0$ difficult?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why is drawing a tangent at $t=0$ difficult?

MODEL ANSWER / EXAMINER EXPECTATION:

Because the curve is steepest at the start, slight errors in drawing the line can lead to large variations in the calculated initial rate.

FAQ #2: How many half-lives must be measured to prove first order?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How many half-lives must be measured to prove first order?

MODEL ANSWER / EXAMINER EXPECTATION:

At least two successive half-lives must be measured to show that they are constant.

FAQ #3: What does a zero order concentration–time graph look like?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What does a zero order concentration–time graph look like?

MODEL ANSWER / EXAMINER EXPECTATION:

It is a straight line with a constant negative gradient.